

Exercise 4: Joins Exercise

Nolwandle Lala Dhladhla

Data Analytics

Bright learn

Date : 13 September 2026

Q U E S T I O N 0 1 Show every user who has a subscription. Match users to subscriptions.

The screenshot shows the Databricks workspace interface. The SQL Editor contains the following query:

```
71 (902,2,703,30),
72 (903,1,702,60),
73 (904,7,701,20),
74 (905,3,705,90),
75
76 SELECT
77   users.user_id,
78   users.user_name,
79   subscriptions.subscription_id,
80   subscriptions.start_date
81 FROM brightlearn_course_students.ed.users
82 INNER JOIN brightlearn_course_students.ed.subscriptions
83   ON users.user_id = subscriptions.subscription_id;
84 SELECT*
85 FROM brightlearn_course_students.ed.users;
86
87
```

The query results are displayed in a table with 5 rows and 3 columns: user_id, user_name, and country.

user_id	user_name	country
1	Nomvula	Johannesburg
2	David	Cape Town
3	Anele	Durban
4	Kabelo	Pretoria
5	Lerato	Pretoria

Q U E S T I O N 0 2 Show every subscription with its matching plan name and monthly price.

The screenshot shows the Databricks workspace interface. The SQL Editor contains the following query:

```
95 (901,1,10,'2026-01-15'),
96 (902,2,11,'2026-02-01'),
97 (903,1,12,'2026-03-10'),
98 (904,6,11,'2026-03-20'),
99 (905,3,13,'2026-04-05');
100
101 SELECT
102   subscriptions.subscription_id,
103   subscriptions.user_id,
104   plans.plan_name,
105   plans.monthly_price
106 FROM brightlearn_course_students.ed.subscriptions
107 INNER JOIN brightlearn_course_students.ed.plans
108   ON subscriptions.plan_id = plans.plan_id;
109
110
111
```

The query results are displayed in a table with 7 rows and 4 columns: subscription_id, user_id, plan_name, and monthly_price.

subscription_id	user_id	plan_name	monthly_price
501	1	Basic	79
502	2	Standard	129
503	1	Premium	199
504	6	Standard	129
505	3	Family	249
501	1	Basic	79
501	2	Standard	129

Q U E S T I O N 0 3 Show every viewing session that has a matching show. Include the show title and genre.

Workspace

Home

Question 1 Exercise 1

Question 1 Exercise 2

Question 2 Exercise 2

Question 3 Exercise 2

Question 4 Exercise 2

Question 5 Exercise 2

Exercise 4

Run all (1000)

1 minute ago (13s)

workspace, brightlearn_course_students

Last edit was now

```

101
102
103 SELECT
104   viewing_sessions.session_id,
105   viewing_sessions.user_id,
106   viewing_sessions.show_id,
107   viewing_sessions.watch_minutes,
108   shows.show_title,
109   shows.genre
110 FROM brightlearn_course_students.e4.viewing_sessions
111 INNER JOIN brightlearn_course_students.e4.shows
112 ON viewing_sessions.show_id = shows.show_id;
113
114
115
116
117

```

Add parameter

Results 16 of 16

Table

	session_id	user_id	show_id	watch_minutes	show_title	genre
1	904	7	701	20	Comedy Hour	Comedy
2	903	1	702	60	Crime Time	Drama
3	902	2	703	30	Tech Tales	Documentary
4	904	7	701	20	Comedy Hour	Comedy
5	903	1	702	60	Crime Time	Drama

QUESTION 4 Show every viewing session with the user who watched it. Only show sessions with a matching user

Workspace

Home

Question 1 Exercise 1

Question 1 Exercise 2

Question 2 Exercise 2

Question 3 Exercise 2

Question 4 Exercise 2

Question 5 Exercise 2

Exercise 4

Run all (1000)

Just now (16s)

workspace, brightlearn_course_students

Last edit was now

```

116 (5, 'Lerato', 'Pretoria');
117
118
119 SELECT
120   users.user_name,
121   users.country,
122   viewing_sessions.session_id,
123   viewing_sessions.show_id,
124   viewing_sessions.watch_minutes
125 FROM brightlearn_course_students.e4.users
126 INNER JOIN brightlearn_course_students.e4.viewing_sessions
127 ON users.user_id = viewing_sessions.user_id;
128
129
130
131
132

```

Add parameter

Results 18 of 18

Table

	user_name	country	session_id	show_id	watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90

QUESTION 5 Show users along with their subscriptions, the plan name, and the price. Use only users who have both a subscription and a valid plan.

Workspace

Home | Question 1 Exercise 1 | Question 1 Exercise 2 | Question 2 Exercise 2 | Question 3 Exercise 2 | Question 4 Exercise 2 | Question 5 Exercise 2 | Exercise 4

Run all (1000) | 1 minute ago (17s) | workspace: brightlearn_course_students | Last edit was now | Edit | Serverless Starte... | Schedule | Save* | Comments | Share

```

118
119 SELECT
120   users.user_name,
121   users.country,
122   plans.plan_name,
123   plans.monthly_price,
124   subscriptions.start_date
125 FROM brightlearn_course_students.e4.users
126 INNER JOIN brightlearn_course_students.e4.subscriptions
127   ON users.user_id = subscriptions.user_ids
128 INNER JOIN brightlearn_course_students.e4.plans
129   ON subscriptions.plan_id = plans.plan_id;
130
131

```

Add parameter

Results 18 of 18 | Table | +

	user_name	country	plan_name	monthly_price	start_date
1	Nomvula	Johannesburg	Basic	79	2026-01-15
2	David	Cape Town	Standard	129	2026-02-01
3	Nomvula	Johannesburg	Premium	199	2026-03-10
4	Anele	Durban	Family	249	2026-04-05
5	Nomvula	Johannesburg	Basic	79	2026-01-15

04 Part B — LEFT JOIN

QUESTION 6 Show every user and any subscriptions they have. Users without subscriptions must still appear.

Workspace

Home | Question 1 Exercise 1 | Question 1 Exercise 2 | Question 2 Exercise 2 | Question 3 Exercise 2 | Question 4 Exercise 2 | Question 5 Exercise 2 | Exercise 4

Run all (1000) | Just now (18s) | workspace: brightlearn_course_students | Last edit was now | Edit | Serverless Starte... | Schedule | Save* | Comments | Share

```

114 ('Anele','Durban'),
115 ('Kabelo','Pretoria'),
116 ('Lerato','Pretoria');
117
118 SELECT
119   users.user_id,
120   users.user_name,
121   subscriptions.subscription_id,
122   subscriptions.start_date
123 FROM brightlearn_course_students.e4.users
124 LEFT JOIN brightlearn_course_students.e4.subscriptions
125   ON users.user_id = subscriptions.user_id;
126
127
128

```

Add parameter

Results 18 of 18 | Table | +

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	4	Kabelo	null	null
5	5	Lerato	null	null

1,000 rows | 17.683s runtime | See performance (18) | Checked for improvements | Refreshed Now

QUESTION 7 Show every plan and the subscriptions on it. Plans with no subscribers must still appear.

The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, and SQL Warehouses. The main area displays a SQL query in the editor, which has been executed. The query is as follows:

```

111 VALUES
112 (1,'Nomvula','Johannesburg'),
113 (2,'David','Cape Town'),
114 (3,'Anele','Durban'),
115 (4,'Kabelo','Pretoria'),
116 (5,'Lerato','Pretoria');
117
118 SELECT
119 plans.plan_id,
120 plans.plan_name,
121 subscriptions.subscription_id,
122 subscriptions.user_id
123 FROM brightlearn_course_students.e4.plans
124 LEFT JOIN brightlearn_course_students.e4.subscriptions
125 ON plans.plan_id = subscriptions.plan_id;

```

Below the query editor, the results are displayed in a table format. The table has 4 columns: `plan_id`, `plan_name`, `subscription_id`, and `user_id`. The results show 5 rows of data:

plan_id	plan_name	subscription_id	user_id
10	Basic	501	1
11	Standard	504	6
12	Premium	503	1
13	Family	505	3
14	Mobile	null	null

The interface also shows a status bar at the bottom indicating '1,000 rows', '17.336s runtime', and 'Checked for improvements'.

QUESTION 08 Show every show and any viewing sessions on it. Shows that were never watched must still appear

The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, and SQL Warehouses. The main area displays a SQL query in the editor, which has been executed. The query is as follows:

```

111 VALUES
112 (1,'Nomvula','Johannesburg'),
113 (2,'David','Cape Town'),
114 (3,'Anele','Durban'),
115 (4,'Kabelo','Pretoria'),
116 (5,'Lerato','Pretoria');
117
118 SELECT
119 shows.show_id,
120 shows.show_title,
121 viewing_sessions.session_id,
122 viewing_sessions.watch_minutes
123 FROM brightlearn_course_students.e4.shows
124 LEFT JOIN brightlearn_course_students.e4.viewing_sessions
125 ON shows.show_id = viewing_sessions.show_id;

```

Below the query editor, the results are displayed in a table format. The table has 4 columns: `show_id`, `show_title`, `session_id`, and `watch_minutes`. The results show 5 rows of data:

show_id	show_title	session_id	watch_minutes
701	Comedy Hour	904	20
702	Crime Time	903	60
703	Tech Tales	902	30
704	Cooking Lab	null	null
706	Wild Earth	null	null

The interface also shows a status bar at the bottom indicating '1,000 rows', '17.336s runtime', and 'Checked for improvements'.

QUESTION 09 Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details).

databricks Search data, notebooks, recents, and more... CTRL + P Verify identity workspace

Workspace Home Question 1 Exercise 1 Question 1 Exercise 2 Question 2 Exercise 2 Question 3 Exercise 2 Question 4 Exercise 2 Question 5 Exercise 2 Exercise 4

Run all (1000) Just now (204) workspace.brightlearn_course_students Last edit was now Edit Serverless Start... 2XS Schedule Save* Comments Share

```
123 FROM brightlearn_course_students.e4.shows
124 LEFT JOIN brightlearn_course_students.e4.viewing_sessions
125 ON shows.show_id = viewing_sessions.show_id;
126
127 SELECT
128   viewing_sessions.session_id,
129   viewing_sessions.show_id,
130   viewing_sessions.watch_minutes,
131   users.user_id,
132   users.user_name
133 FROM brightlearn_course_students.e4.viewing_sessions
134 LEFT JOIN brightlearn_course_students.e4.users
135 ON viewing_sessions.user_id = users.user_id;
136
137
```

Add parameter

Results 19 of 19 Table +

	session_id	show_id	watch_minutes	user_id	user_name
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele
6	901	701	45	1	Nomvula

QUESTION 10 Show every user, the plan they are on (if any), and the monthly price. Users without a subscription must still appear.

Workspace

Home

Run all (1000) ✓ 1 minute ago (23s) workspace.brightlearn_course_students

```

133 FROM brightlearn_course_students.e4.viewing_sessions
134 LEFT JOIN brightlearn_course_students.e4.users
135 ON viewing_sessions.user_id = users.user_id;
136
137 SELECT
138   users.user_name,
139   users.country,
140   plans.plan_name,
141   plans.monthly_price
142 FROM brightlearn_course_students.e4.users
143 LEFT JOIN brightlearn_course_students.e4.subscriptions
144 ON users.user_id = subscriptions.user_id
145 LEFT JOIN brightlearn_course_students.e4.plans
146 ON subscriptions.plan_id = plans.plan_id;
147

```

Results 20 of 20

	user_name	country	plan_name	monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Pretoria	null	null

1,000 rows 22.626s runtime See performance (20) ✓ Checked for improvements Refreshed Now

05 Part C — FULL OUTER JOIN

QUESTION 1 Show every user and every subscription, including users without subscriptions AND subscriptions referencing users that do not exist.

Workspace

Home

Run all (1000) ✓ Just now (26s) workspace.brightlearn_course_students

```

144 ON users.user_id = subscriptions.user_id
145 LEFT JOIN brightlearn_course_students.e4.plans
146 ON subscriptions.plan_id = plans.plan_id;
147
148 SELECT
149   users.user_id,
150   users.user_name,
151   subscriptions.subscription_id,
152   subscriptions.start_date
153 FROM brightlearn_course_students.e4.users
154 FULL OUTER JOIN brightlearn_course_students.e4.subscriptions
155 ON users.user_id = subscriptions.user_id;
156
157
158

```

Results 21 of 21

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15
5	1	Nomvula	503	2026-03-10

QUESTION 12 Show every plan and every subscription, including plans without subscribers AND any subscription referencing a plan that does not exist.

The screenshot shows the Databricks SQL Editor interface. The query is as follows:

```

202 (503,1,12,'2026-03-10'),
203 (504,6,11,'2026-03-20'),
204 (505,3,13,'2026-04-05');
205
206 SELECT
207     plans.plan_id,
208     plans.plan_name,
209     subscriptions.subscription_id,
210     subscriptions.user_id
211 FROM brightlearn_course_students.e4.plans
212 FULL OUTER JOIN brightlearn_course_students.e4.subscriptions
213     ON plans.plan_id = subscriptions.plan_id;
  
```

The results table shows 6 rows of data:

	plan_id	plan_name	subscription_id	user_id
1	12	Premium	503	1
2	10	Basic	501	1
3	13	Family	505	3
4	14	Mobile	null	null
5	11	Standard	504	6
6	11	Standard	502	2

The interface also shows a sidebar with navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, and SQL Warehouses. The bottom status bar indicates 6 rows, 34.906s runtime, and a checked status for improvements.

QUESTION 13 Show every show and every viewing session, including shows that were never watched AND sessions referencing shows that do not exist.

The screenshot shows the Databricks SQL Editor interface. The query is as follows:

```

256 (905,3,705,90);
257
258 SELECT
259     shows.show_id,
260     shows.show_title,
261     viewing_sessions.session_id,
262     viewing_sessions.watch_minutes
263 FROM brightlearn_course_students.e4.shows
264 FULL OUTER JOIN brightlearn_course_students.e4.viewing_sessions
265     ON shows.show_id = viewing_sessions.show_id;
266
267
  
```

The results table shows 7 rows of data:

	show_id	show_title	session_id	watch_minutes
1	704	Cooking Lab	null	null
2	703	Tech Tales	902	30
3	701	Comedy Hour	904	20
4	702	Crime Time	903	60
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45
7	null	null	905	90

The interface also shows a sidebar with navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL Editor, Queries, Dashboards, Genie Agents, Alerts, Query History, and SQL Warehouses. The bottom status bar indicates 7 rows, 40.643s runtime, and a checked status for improvements.

QUESTION 14 Show every user and every viewing session, including users with no sessions AND sessions referencing users who do not exist.

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

```
266 ON shows.show_id = viewing_sessions.show_id;
267
268 SELECT
269     users.user_id,
270     users.user_name,
271     viewing_sessions.session_id,
272     viewing_sessions.show_id,
273     viewing_sessions.watch_minutes
274 FROM brightlearn_course_students.e4.users
275 FULL OUTER JOIN brightlearn_course_students.e4.viewing_sessions
276 ON users.user_id = viewing_sessions.user_id;
277
```

Below the query editor, the results are displayed in a table with 7 rows and 6 columns:

	user_id	user_name	session_id	show_id	watch_minutes
1	1	Nomvula	903	702	60
2	3	Anele	905	705	90
3	1	Nomvula	903	702	60
4	3	Anele	905	705	90
5	1	Nomvula	903	702	60
6	3	Anele	905	705	90
7	1	Nomvula	903	702	60

QUESTION 15 Show every user, every subscription, and every plan in one query — using FULL OUTER JOIN throughout. This is the hardest question — get all gaps visible at once.

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

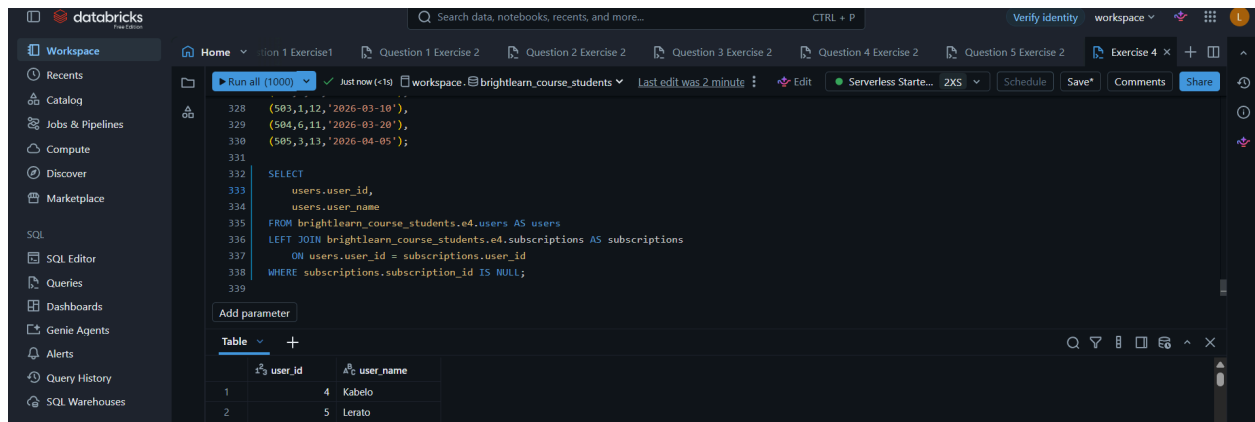
```
296 SELECT
297     users.user_id,
298     users.user_name,
299     subscriptions.subscription_id,
300     plans.plan_id,
301     plans.plan_name
302 FROM brightlearn_course_students.e4.users AS users
303 FULL OUTER JOIN brightlearn_course_students.e4.subscriptions AS subscriptions
304 ON users.user_id = subscriptions.user_id
305 FULL OUTER JOIN brightlearn_course_students.e4.plans AS plans
306 ON subscriptions.plan_id = plans.plan_id;
307
```

Below the query editor, the results are displayed in a table with 7 rows and 6 columns:

	user_id	user_name	subscription_id	plan_id	plan_name
1	1	Nomvula	503	12	Premium
2	1	Nomvula	503	12	Premium
3	1	Nomvula	503	12	Premium
4	1	Nomvula	503	12	Premium
5	1	Nomvula	503	12	Premium
6	1	Nomvula	503	12	Premium
7	1	Nomvula	503	12	Premium

06 Bonus Challenge Now the conceptual test.

B O N U S 0 1 Which users have not subscribed to any plan?



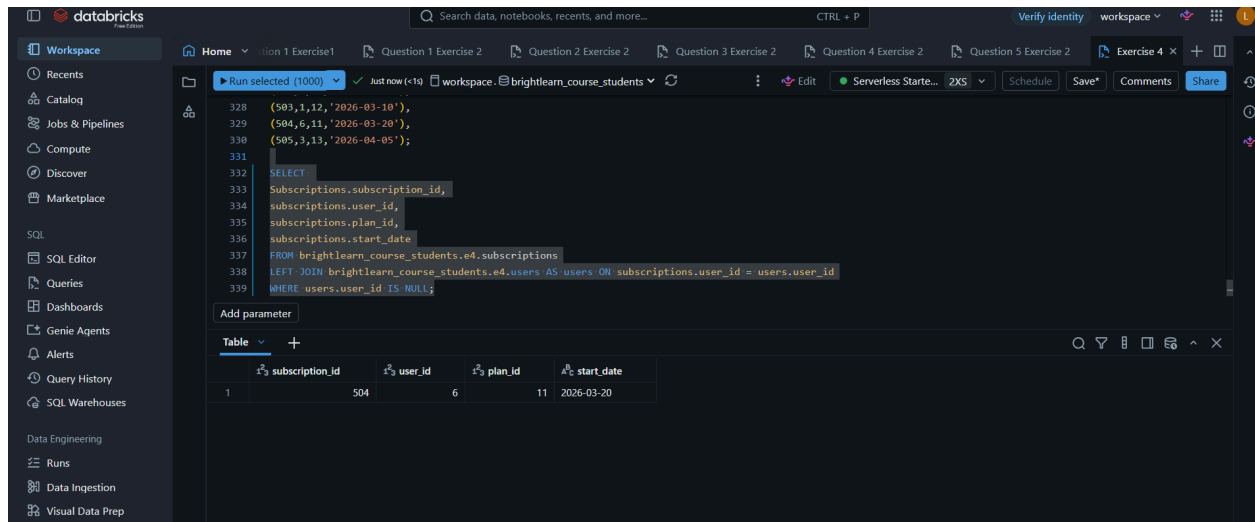
The screenshot shows the Databricks workspace interface. The SQL Editor contains the following query:

```
328 (503,1,12,'2026-03-10'),
329 (504,6,11,'2026-03-20'),
330 (505,3,13,'2026-04-05');
331
332 SELECT
333     users.user_id,
334     users.user_name
335 FROM brightlearn_course_students.e4.users AS users
336 LEFT JOIN brightlearn_course_students.e4.subscriptions AS subscriptions
337     ON users.user_id = subscriptions.user_id
338 WHERE subscriptions.subscription_id IS NULL;
339
```

The results table shows the following data:

user_id	user_name
4	Kabelo
5	Lerato

B O N U S 0 2 Which subscriptions reference users that do not exist in the users table?



The screenshot shows the Databricks workspace interface. The SQL Editor contains the following query:

```
328 (503,1,12,'2026-03-10'),
329 (504,6,11,'2026-03-20'),
330 (505,3,13,'2026-04-05');
331
332 SELECT
333     subscriptions.subscription_id,
334     subscriptions.user_id,
335     subscriptions.plan_id,
336     subscriptions.start_date
337 FROM brightlearn_course_students.e4.subscriptions
338 LEFT JOIN brightlearn_course_students.e4.users AS users ON subscriptions.user_id = users.user_id
339 WHERE users.user_id IS NULL;
```

The results table shows the following data:

subscription_id	user_id	plan_id	start_date
504	6	11	2026-03-20

B O N U S 0 3 Which shows have never been watched?

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

```
345 (984, 7, 791, 29),
346 (985, 3, 795, 98);
347
348 SELECT
349 shows.show_id,
350 shows.show_title,
351 shows.genre
352 FROM brightlearn_course_students.e4.shows
353 LEFT JOIN brightlearn_course_students.e4.viewing_sessions v ON shows.show_id = v.show_id
354 WHERE v.show_id IS NULL;
```

The results table below the query shows 6 rows of data:

	show_id	show_title	genre
1	704	Cooking Lab	Lifestyle
2	706	Wild Earth	Documentary
3	704	Cooking Lab	Lifestyle
4	706	Wild Earth	Documentary
5	704	Cooking Lab	Lifestyle
6	706	Wild Earth	Documentary

B O N U S 0 4 Which viewing sessions reference shows that do not exist?

The screenshot shows the Databricks SQL Editor interface. The query editor contains the following SQL code:

```
346 (985, 3, 795, 98);
347
348 SELECT
349 viewing_sessions.session_id,
350 viewing_sessions.user_id,
351 viewing_sessions.show_id,
352 viewing_sessions.watch_minutes
353 FROM brightlearn_course_students.e4.viewing_sessions
354 LEFT JOIN brightlearn_course_students.e4.shows ON brightlearn_course_students.e4.viewing_sessions.show_id = shows.show_id
355 WHERE shows.show_id IS NULL;
```

The results table below the query shows 2 rows of data:

	session_id	user_id	show_id	watch_minutes	
1		905	3	705	90
2		905	3	705	90

B O N U S 0 5 Which plans have no subscribers?

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Discover

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie Agents

Alerts

Query History

SQL Warehouses

Data Engineering

Runs

Data Ingestion

Home

Question 1 Exercise1

Question 1 Exercise 2

Question 2 Exercise 2

Question 3 Exercise 2

Question 4 Exercise 2

Question 5 Exercise 2

Exercise 4

Run selected (1000)

Just now (<1s)

workspace: brightlearn_course_students

Last edit was no

Edit

Serverless Starte...

2XS

Schedule

Save*

Comments

Share

345 (904,7,701,20),

346 (905,3,705,90),

347

348 SELECT

349 plans.plan_id,

350 plans.plan_name,

351 plans.monthly_price

352 FROM brightlearn_course_students.e4.plans AS plans

353 LEFT JOIN brightlearn_course_students.e4.subscriptions AS subscriptions_id ON plans.plan_id = subscriptions_id.plan_id

354 WHERE subscriptions_id.plan_id IS NULL;

355

356

Add parameter

Table

plan_id

plan_name

monthly...

plan_id

plan_name

monthly...

plan_id

plan_name

monthly...

plan_id	plan_name	monthly...
1	14 Mobile	59
2	14 Mobile	59